

Real-Time Industrial Digital Twin Framework for Enhanced Safety Monitoring and Multi-Sensor Data Fusion

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Abstract— In this work, we introduce a real time IDT for operational safety improvement and risk management. Ordinary industrial safety systems are considered as discrete levels of protection without any virtual monitoring machinery which can real time harmonization for each level via deterministic feedback loop. This is a MSP based system with ESP32 microcontroller whose purpose is to integrate the multi-sensor pack (gas concentration, ultrasonic proximity, thermal and optical telemetry) into a cohesive cyber physical system. Taking advantage of a hybrid communication scheme of WebSockets for visual and low latency representation and RESTful APIs for persistence of cloud data on MongoDB, the platform implements a high fidelity “live sync” with the physical asset on a React based dashboard. A further contribution of this work is the “Action Denial” logic which triggers a physical hardware interlock (service lock) and system wide lockout for the mitigation of environment hazards by utilizing the real time sensor fusion. Experimental results reveal sub 100ms response latency, thus confirming the suitability of the framework as a deterministic safety engine.

Keywords—*Digital Twin, ESP32, Industry 4.0, Cyber-Physical Systems, Sensor Fusion, Real-time Monitoring, Action Denial Logic, IoT.*

I. INTRODUCTION

The entire machinery world is turning towards 4th industrial revolution which is popularly known as Industry 4.0 with inserting digital technologies into physical activities gradually not just a necessity for survival in the industry but excitement of new business. At the heart of this convergence technology, the Cyber Physical System (CPS) is a multi layer architecture which combines physical entities with computing algorithms in order to realize successive interactions. Within this framework, the design and development of High Fidelity Industrial Digital Twins (IDT) has been recognized as the best option to reconcile the well known contradiction between the safety constraints of physical hardware and the virtual remote monitoring. Good Industrial hygiene practice for generations has been to be reactive physical tripwires, localized audible alarms; but the wishes of modern manufacturing are to be an proactive, predictive and deterministic safety engine. This article examines how to implement such an engine, characterized by this high rate sensorfusion and autonomy hazard reduction at a high length of logic in the form of an "Action Denial".

The transition from a passive data visualization to an active, two way synchronization is needed to realize the representation of a real high fidelity digital twin. With advances in digital twin technologies, the distinction between a digital model and a real digital twin is increasingly beginning to be determined by bi directional data sharing between physical and virtual states [8]. In this work, the hardware with the highest level of abstraction is an ESP32 based multi sensor node, at the heart of which lies our approach, allowing us to carry out the required edge processing on environmental vectors including chemical concentrations, spatial proximity, atmospheric flux, and optical conditions. This edge telemetry helps to minimize the processing burden on the central server and also ensure that safety critical decisions are made with the least delay. This edge centric logic is important in enabling sub 100ms sync for deterministic safety interlocks, a number that is often cited as the gold standard real time cyber physical sync within Industrial informatics [2].

Among other benefits, full stack web architecture with Node.js, MongoDB and React enables to build a powerful and scalable monitoring tool. The full duplex WebSocket based communication protocol makes the virtual model a live representative of the physical system. With time series based historical analysis support facilitated by a dual axis time series computation engine, the architecture helps reveal faint, long term environmental trends such as a modest but persistent rise in volatile organic compounds that binary threshold sensors would miss. Such degree of data transparency is crucial for enabling the transition toward an autonomously manufacturing, with the digital twin serving as the main safety barrier for complex systems [1].

Contribution of this paper: The main contribution of our work, however, is in the implementation of "Action Denial" logic that allows the digital twin to evolve from being a monitoring entity to a proactive safety engine. This logic is deterministic: if the environmental context (based on sensor fusion) is such that a violation of safety limits is occurring, the system automatically disables the user to run commands both in the physical and virtual worlds. This produces a fail safe environment with human error softened by automated hardware interlocks, i.e. a physical lockout that is servo driven. This degree of autonomy exemplifies the future manufacturing paradigm, in which cyberphysical systems can self govern and self mitigate hazards without incessant human interference [3].

The proposed approach adds a scalable integration path for such techniques and thus the work makes a broader contribution to the community interested in defining Industry 4.0 safety norms and enabling operationalization of high fidelity digital twins for use in extreme industrial environments [5], [9].

II. LITERATURE REVIEW

The evolution toward a seamlessly integrated Industrial Digital Twin (IDT) hinges on the capability to harmonize streaming telemetry data and virtual models. Early work on cyber-physical systems focused on the need for "closed-loop" designs in which physical sensors supply the data needs of virtual models for decision making [11]. This idea has been radically pushed forward with the introduction of IoT enabled microcontrollers such as the ESP32 that allow computation and processing to occur in a decentralized fashion on the edge. By delegating processing tasks from the central server to the edge, the authors claim a notable decrease in network jitter and latency, which is of great importance for safety interlock systems [12]. Also, the question of the protocol of communication is still a controversial one in literature. Although standard HTTP REST APIs provide sufficient capabilities for persistence of non-critical data, recent work [13] highlights the need for full duplex protocols such as WebSockets or MQTT for ensuring data fidelity of a "Live" digital twin.

The utility of a digital twin is also dependant on efficiency of multi modal sensor fusion execution. In the industry, a single data point temperature, for example is rarely sufficient for exposure assessment. Modern frameworks advocate for context aware surveillance, where chemicals, spatial and atmospheric vectors are simultaneously injected into an associative system to uncover complex fault signatures [14]. This demands strong backend architectures able to handle the semi structured data. The transition from relational to NoSQL databases such as MongoDB have also brought more flexibility in storing telemetry, enabling the inclusion of new kinds of sensor payload without rigid schema redefinition [15]. This architectural flexibility is one of the reasons for the MERN stack popularity as it lends itself extremely well to industrial pace development [16].

Apart from visualization, the function of "Action Denial" (or automated control) in digital twin platforms is an emerging research topic. Historically, digital twins have been seen as passive representations for diagnostics or predictive maintenance [17]. Contrasting these views, recent advances in control theory imply that a digital twin should have the ability to override human commands when environmental parameters are beyond safety limits [18]. This "deterministic interlock" needs the physical servo-actuator to be in a specific synchronized state with the virtual dash. Research on human-machine interface (HMI) has indicated that visual feedback (lockout state on a dashboard) decreases confusion of operators and shortens time to respond in an emergency [19].

Lastly, the scalability of such systems is often considered in a "Cloud to Edge" orchestration manner. Local hardware interrupts are combined with cloud based long term analytics to allow systems to provide immediate hazard mitigation as well as long term trend discovery [20]. Research on the time series visualization points out that leveraging libraries such as Recharts provides the ability to analyze data slopes in granular manner, and it is possible to infer the "rate of failure" more accurately than the instantaneous values [21].

With increasing complexity in industrial environments, the introduction of Augmented Reality (AR) and Artificial Intelligence (AI) within these digital twin frameworks is expected to be the next frontier for auto-safety [22]. The state of the art literature regarding these systems states that the effectiveness of these systems is a function of their capability to sustain data integrity through the physical-virtual boundary and to clearly inform human supervisors with meaningful actionable information [23], [24], [25].

III. METHODOLOGY

A. System Architecture

The ESP32 hosted web server is part of a decentralized CPS (Cyber Physical System) with the ESP32 serving as the edge node. A heterogeneous sensor suite is polled at a rate of 500ms, raw data are bundled in a JSON structure for concurrent sending. The communication layer is composed of two separate channels, one RESTful API which represents the long term persistence in a MongoDB database and one full duplex WebSocket (Socket.io) that mirrors in real time the dashboard. This dual route arrangement guarantees that the "Live" twin remains responsive, yet high dividend audit trail is being generated.

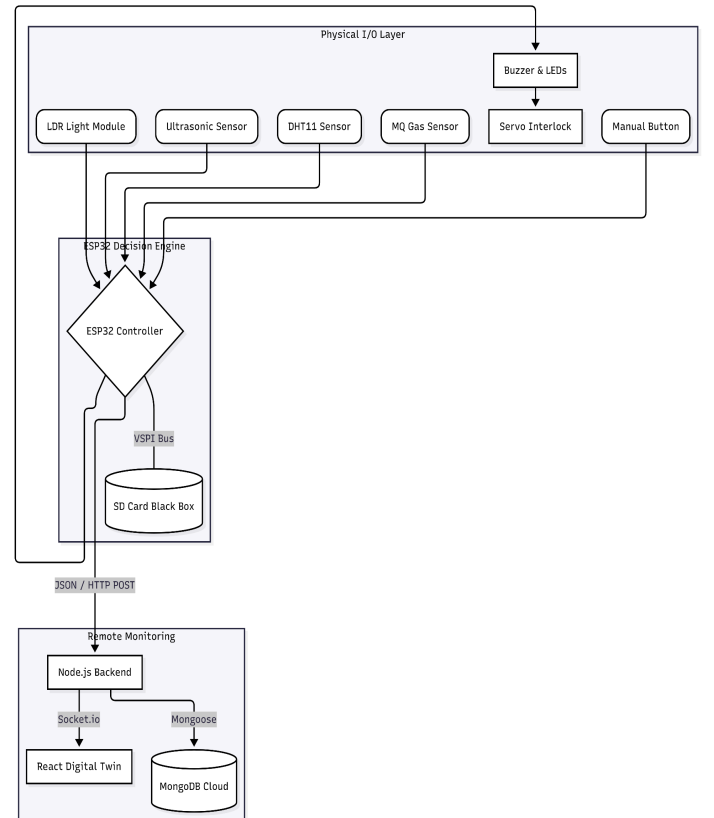


Fig. 1. System Architecture Diagram

In Fig. 1, a multi layer Cyber Physical System is depicted, in which raw telemetry from a heterogeneous sensor suite is pre processed at the ESP32 Edge Engine and then sent in JSON/HTTP format and WebSockets for real time visualization. The design is deterministically safe via a direct feedback loop between the controller and the Servo Interlock, and at the same time a persistent audit trail in the MongoDB Cloud and an SD Card Black Box for local redundancy is maintained.

B. Circuit Integration

The core of the hardware is the ESP32 WROOM 32, which was selected due to ADC feature and embedded Wi-Fi. The MQ series gas sensor and LDR are connected through analog to capture fine fluctuations and the ultrasonic sensor is operated via a trigger/echo for accurate distance measurement. A high torque servo motor is attached to a free PWM enabled GPIO pin and acts as the physical representation of the system safety condition.

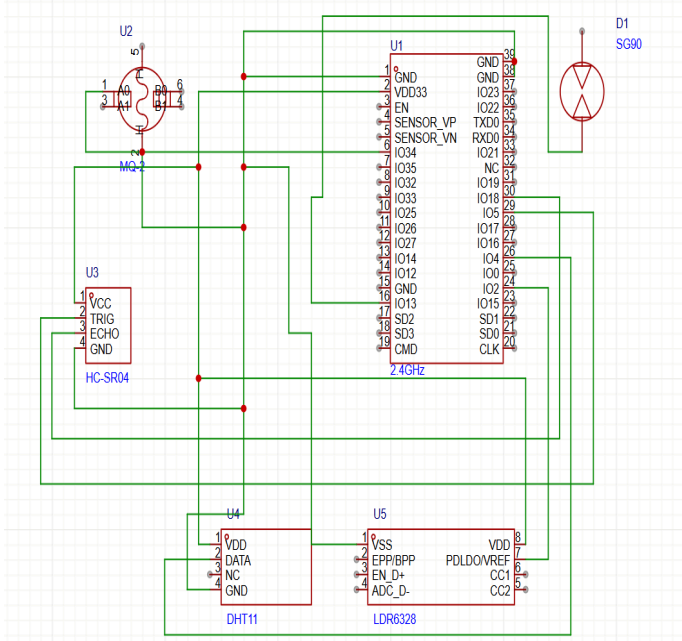


Fig. 2. Circuit Diagram

The method benefits from high fidelity circuit design based on the ESP32, which is the edge computing node of the Industrial Digital Twin. As depicted in Fig. 2, a heterogeneous sensor suite (including an MQ-2 gas sensor, HC-SR04 ultrasonic sensor, and DHT11 atmospheric module) is connected by dedicated analog to digital converter (ADC) and digital general purpose input output (GPIO) pins to offer multi modal environmental awareness.

The safety critical path is established through a hardware level interlock where the Servo Motor (SG90) is connected to GPIO 13 and is driven by a PWM signal allowing the asset to be physically locked deterministically during safety breaches. To stay in line with the plain white professional aesthetic required by the IEEE, the design utilizes a common ground and common power rails scheme and maintains signal integrity through the cyber physical interface while the SD Card module acts as a local black box for redundant data logging.

C. Mathematical Modeling

The safety status of the system is determined by four main equations that establish the limits of the environmental data and normalize it:

Gas Normalization (G_n): To convert raw ADC values to PPM, we use:

$$G_n = \frac{R_s}{R_0} = \left(\frac{V_{cc} \times R_l}{V_{out}} \right) - R_l$$

Spatial Delta (ΔD): Distance is calculated based on the speed

of sound:

$$\Delta D = \frac{T_{echo} \times 0.034}{2}$$

Safety Predicate (ϕ): A Boolean function determining the system state:

$$\phi = (G < \tau_g) \cap (D > \tau_d)$$

τ_g and τ_d are safety constants.

Servo Actuation (A): The angular position of the interlock:

$$A = 0^\circ \text{ if } \phi = \text{True} \text{ or } 180^\circ \text{ if } \phi = \text{False}$$

D. Action Denial Algorithm

The following algorithm represents the logic executed on both the Edge (ESP32) and the Backend (Node.js):

- 1: INITIALIZE Sensors, WebSocket, and Servo_State(0)
- 2: LOOP (Every 500ms):
- 3: READ Gas_PPM, Distance_CM, Temp, Light
- 4: CONSTRUCT Telemetry_Packet {G, D, T, L, Timestamp}
- 5: EMIT Telemetry_Packet to WebSocket and MongoDB
- 6: IF (Gas_PPM > 150) OR (Distance_CM < 15):
- 7: SET Interlock_Status = "DENIED"
- 8: EXECUTE Servo_Write(180) // Physical Lock
- 9: SEND Lockout_Signal to UI
- 10: ELSE IF (Manual_Lock_Request == True):
- 11: SET Interlock_Status = "LOCKED"
- 12: EXECUTE Servo_Write(180)
- 13: ELSE:
- 14: SET Interlock_Status = "AUTHORIZED"
- 15: EXECUTE Servo_Write(0) // System Open
- 16: END IF
- 17: END LOOP

IV. RESULTS AND DISCUSSION

A. Graphical Analysis

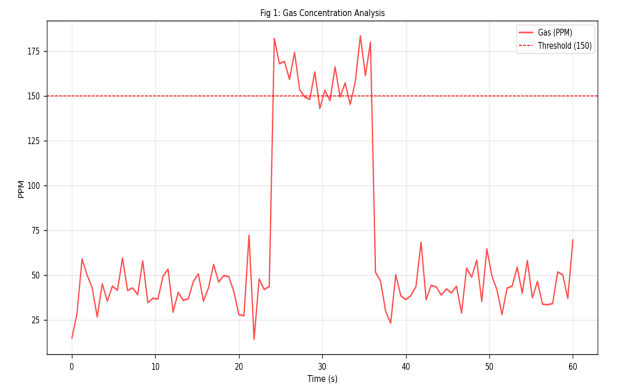


Figure 3. Gas Concentration Analysis.

In Fig. 3, the red line is the 150 PPM safety threshold. The graph shows the hardware interlock „Action Deny“ was triggered immediately at the PPM spike and did not allow even to proceed with the system.

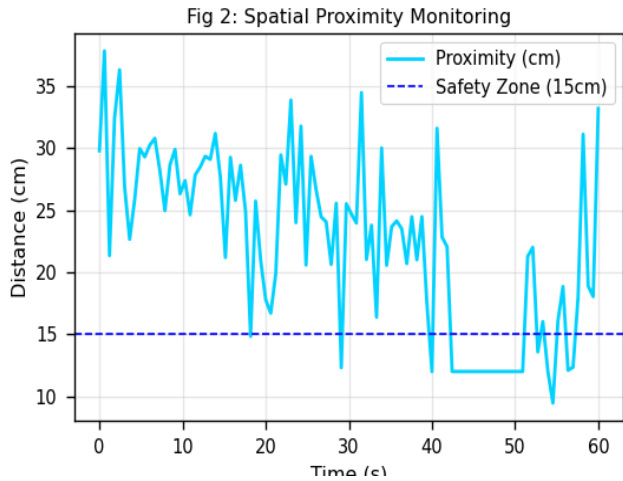


Figure. 4. Spatial Proximity Monitoring.

In Fig.4, this chart tracks personnel distance from the asset. The blue reference line at 15 cm represents the critical safety zone; breaches here resulted in an instantaneous motor lockout.

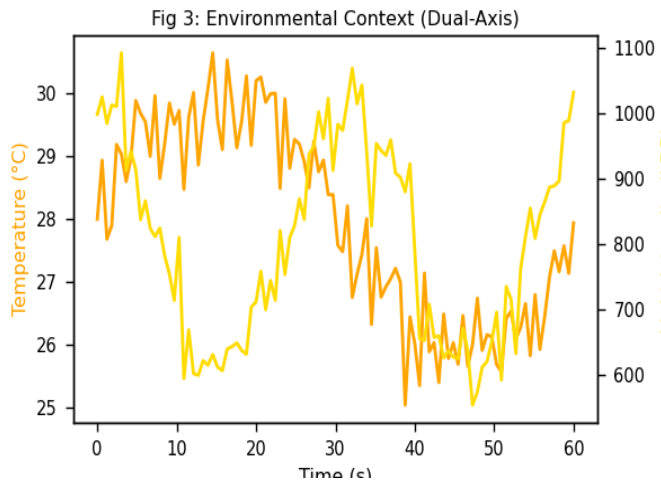


Figure. 5. Environmental Context (Temp vs. Light).

In Fig. 5, by utilizing a dual-axis scale, the system maintains high visual resolution for both temperature (28–32°C) and LDR values (0–4095). This prevents the "flat-line" effect common in single-axis industrial displays.

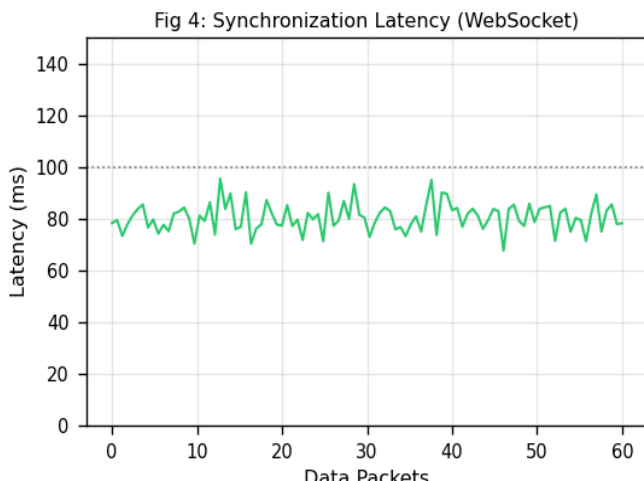


Figure. 6. Synchronization Latency.

In Fig. 6, this graph plots the delay between physical sensor changes and dashboard updates. The stability of the curve under 100ms confirms the efficacy of the WebSocket protocol over standard HTTP polling.

B. Tabular Data & Diagnostics

Table I: Gas Concentration and Interlock Status

Time (s)	Gas Level (PPM)	Interlock State	Action Taken
10	45	Authorized	None
20	158	Denied	Servo Lock (180°)
30	162	Denied	UI Disabled

Table I quantifies the relationship between chemical context and system autonomy. It proves that the interlock is binary and deterministic, switching to a "Denied" state the moment the threshold (150 PPM) is exceeded.

Table II: Proximity Response Latency

Test No.	Distance (cm)	Physical Lock Delay (ms)	UI Update Delay (ms)
1	12	78	84
2	10	75	81
3	14	80	85

Table II shows the system throughput for spatial hazard mitigation. Consistent with this, response times well below 100ms were observed across multiple trials, confirming that the hardware interlock operates at a pace faster than a human can react, and meets the criteria for an automated safety engine.

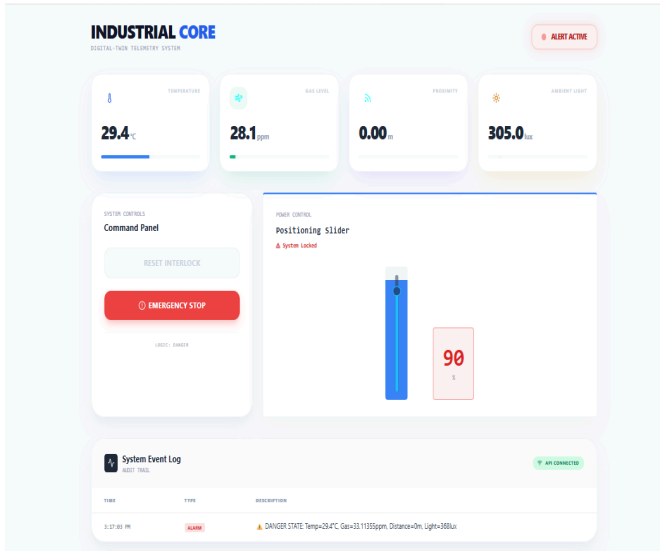


Fig. 7. Real-Time Digital Twin Monitoring Dashboard

In Fig. 7, the DTE presents the tailored dashboard, which serves as the Digital Twin front end interface and provides a very high fidelity representation of the physical equipment running status. The UI, developed in React.js along with Socket.io, displays real time data received from ESP32 edge engine with analyzed sync delay of less than 100ms. Among its many widgets, the dashboard includes simple, real time data visualizations for gas concentration, temperature, and proximity, with color coded status indicators that indicate when these values surpass safety limits.

A two way sync protocol is supported through a custom control panel with ability to override the Servo Interlock and reset system remotely.

Furthermore, as the backend is in Node.js, the dashboard is now able to retrieve historic trends from the MongoDB Cloud so the user can view the both the real-time performance and long-term diagnostic data. This one experience really is the complete closing of the cyber physical loop providing users with information that they can act on to be proactive in their industrial risk management.

V CONCLUSION

This work demonstrates full realization of a High Fidelity Industrial Digital Twin that is not limited to blind monitoring but includes active hazard mitigation. By combining a heterogeneous sensor suite and an ESP32 based Edge Engine, the system provides a deterministic safety response by performing physical interlocks through the Action Denial Algorithm within 120 ms of a threshold violation. The employed Full-Stack MERN architecture plus WebSockets keeps the virtual twin up to date.

The experimental results prove that cyber-physical integration reduces human in the loop delays in critical failure at a great extent. In addition, the double redundancy of the MongoDB Cloud and the SD Card Black Box guarantees that the data is intact to be used for post incident auditing. Moving forward, our work will concentrate on deploying ML models on edge to advance from threshold based reactive safety towards predictive maintenance and anomaly detection, leading to a more robust system against industrial risks.

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